MATH 110 — Week 1

Homework 1 — Solutions

HW1 P1(a): images of inverse images. **Idea.** Membership in the inverse image already requires the output to lie in V.

1. If $y \in f(f^{-1}(V))$, then $y = f(x)$ for some x with $f(x) \in V$. Hence $y \in V$.
2. More precisely, $f(f^{-1}(V)) = V \cap f(X)$: the forward inclusion follows above and from being an output; for the reverse, if $y \in V \cap f(X)$, choose x with $f(x) = y$, so $x \in f^{-1}(V)$.
3. Equality with V therefore holds exactly when $V \subseteq f(X)$. It holds for every $V \subseteq Y$ exactly when f is surjective (take $V = Y$ for necessity).

HW1 P1(b): inverse images of images. **Idea.** Each element of U supplies its own witness that its output belongs to $f(U)$.

1. If $x \in U$, then $f(x) \in f(U)$, hence $x \in f^{-1}(f(U))$.
2. Equality holds exactly when there is no $x \notin U$ whose value equals the value at some $u \in U$; equivalently, $f(U) \cap f(X \setminus U) = \varnothing$.
3. This says U is a union of complete *fibers*, where a fiber means a set $f^{-1}(\{y\})$ of all inputs with one fixed value.
4. If f is injective, $f(x) = f(u)$ implies $x = u$, giving equality for every U. Conversely, equality for every singleton $U = \{u\}$ forces any x with $f(x) = f(u)$ to equal u, proving injectivity.

HW1 P1(c): strict examples and the quantifiers. **Idea.** A missed output makes (a) strict; a shared output across U and its complement makes (b) strict.

1. Let $X = \{0\}$, $Y = \{0, 1\}$, $f(0) = 0$, and $V = \{1\}$. Then $f(f^{-1}(V)) = \varnothing \subsetneq \{1\}$.
2. Let $X = \{0, 1\}$, $Y = \{0\}$, $f(0) = f(1) = 0$, and $U = \{0\}$. Then $f^{-1}(f(U)) = \{0, 1\} \supsetneq U$.
3. Global surjectivity or injectivity is not necessary for a *particular* subset: $V = \varnothing$ always gives equality in (a), and $U = X$ always gives equality in (b). The exact fixed-subset conditions are those proved above.

HW1 P2 (LADR 1A.7). **Idea.** The square is the conjugate, so the cube becomes a real sum of squares.

1. Set $z = (-1 + \sqrt{3}\, i)/2$. Using $i^2 = -1$,
$$z^2 = \frac{1 - 2\sqrt{3}\, i - 3}{4} = \frac{-1 - \sqrt{3}\, i}{2}.$$
2. Multiply: $z^3 = \frac{(-1-\sqrt{3}\, i)(-1+\sqrt{3}\, i)}{4} = \frac{1+3}{4} = 1$. This is precisely the definition of a cube root of 1.

HW1 P3. **Idea.** A nonzero scalar can be removed using its inverse.

1. If $a = 0$, the claimed alternative already holds.
2. Otherwise a^{-1} exists by the field axiom, and $b = (a^{-1}a)b = a^{-1}(ab) = a^{-1}0 = 0$, using associativity and the identity $c0 = 0$ proved in the handout.

HW1 P4 (LADR 1B.2). **Idea.** Use the same inverse argument with scalar multiplication of a vector.

1. If $a = 0$, the assertion holds.
2. If $a \neq 0$, then $v = 1v = (a^{-1}a)v = a^{-1}(av) = a^{-1}0_V = 0_V$, by the vector-space axioms and Notes Proposition 4.

HW1 P5 (LADR 1C.4). **Idea.** The zero function forces the right-hand side to be zero; integration preserves sums and scalar multiples.

1. A continuous function on $[0, 1]$ has no discontinuity there; its definite integral $\int_0^1 f$ is the usual signed area, defined by the limit of Riemann sums. We use the elementary-calculus facts that continuous functions are integrable, expressions $af + g$ formed from continuous f, g and real a are continuous, and $\int(af + g) = a\int f + \int g$.
2. If U_b is a subspace, it contains the zero function. Its integral is 0, so the defining equality forces $b = 0$.
3. Suppose $b = 0$. The zero function belongs to U_0. For $f, g \in U_0$, $a \in \mathbb{R}$, $af + g$ is continuous and
$$\int_0^1 (af + g) = a\int_0^1 f + \int_0^1 g = a \cdot 0 + 0 = 0.$$
4. Thus $af + g \in U_0$; the subspace test proves the result. For example $f(x) = x - 1/2$ belongs to U_0 since $\int_0^1 (x - 1/2)\,dx = 0$.

HW1 P6 (LADR 1C.5). **Idea.** Closure must hold for every scalar in the specified field, including i.

1. $(1, 0) \in \mathbb{R}^2$, and i is an allowed scalar in the complex vector space $\mathbb{C}^2$.
2. $i(1, 0) = (i, 0) \notin \mathbb{R}^2$. Scalar closure fails, so the answer is no.

HW1 P7. **Idea.** Repeated squaring eliminates the cross term because $1 + 1 = 0$ in $\mathbb{F}_2$.

1. In the polynomial ring over $\mathbb{F}_2$ (finite sums of formal powers of x with coefficients in $\mathbb{F}_2$), $(x + 1)^2 = x^2 + 2x + 1 = x^2 + 1$.
2. Square again: $(x + 1)^4 = (x^2 + 1)^2 = x^4 + 1$.
3. Square once more: $(x + 1)^8 = (x^4 + 1)^2 = x^8 + 1$.
4. Over $\mathbb{R}$, the expansion instead is
$$(x + 1)^8 = x^8 + 8x^7 + 28x^6 + 56x^5 + 70x^4 + 56x^3 + 28x^2 + 8x + 1.$$
All seven intermediate coefficients are even, so reduce to zero modulo 2. Over $\mathbb{R}$ they remain nonzero.

Exam practice: Midterm 1 review, Section II, Exercise 7. **Idea.** Intersection preserves both subspace conditions at once; union can fail to contain a sum.

1. If U, W are subspaces, both contain 0, so $0 \in U \cap W$. If $u, v \in U \cap W$ and $a \in F$, closure in each subspace gives $au + v \in U \cap W$.
2. In $\mathbb{R}^2$, let $U = \{(t, 0) : t \in \mathbb{R}\}$ and $W = \{(0, t) : t \in \mathbb{R}\}$. Each is a subspace by direct substitution in the subspace test.
3. $(1, 0), (0, 1) \in U \cup W$ but their sum $(1, 1)$ is in neither axis. Thus a union need not be a subspace.